Dual-Alignment Domain Adaptation for Pedestrian Trajectory Prediction

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***Abstract*—Predicting the plausible future paths of pedestrians is essential for human-involved applications (e.g., autonomous driving and service robotics). Existing pedestrian trajectory prediction methods mainly focus on the performance of multi-scene trained models in single-scene tests, neglecting the cross-scene knowledge differences in practice. To address this issue, we propose a generic dual-alignment framework for pedestrian trajectory prediction. Concretely, we analyze the domain difference at macro and micro scales and mitigate them respectively: at macro scale, an attention-based temporal convolutional generative model transfers the paths of pedestrians and their interaction information from the source domain to the target domain to align the data-level distributions; at micro scale, an auxiliary adversarial network is integrated to assist in training the prediction network to align the feature-level domain-invariant knowledge. Cross-domain experiments demonstrate that our approach significantly improves the performance of existing pedestrian trajectory prediction benchmarks (up to 53.5%) and outperforms previous domain adaptive works (up to 41.7%).**

***Index Terms*—Deep learning methods, human and humanoid motion analysis and synthesis, social HRI, pedestrian trajectory prediction.**

I. Introduction

PEDESTRIAN trajectory prediction attempts to predict future paths by leveraging observed trajectory segments. As a key module in human-robot interaction systems such as self-driving vehicles [1] and autonomous navigation robots [2], it involves modeling and inference about human behaviors in various scenarios to enhance the safety and intelligence of real-world ego-agents [3].

Due to the complex mutual influence between pedestrians and human decision-making's intrinsic uncertainty, predicting human trajectories remains a huge challenge [4]. In recent years, some methods based on recurrent neural networks (RNNs) [5], [6], [7] and graph neural networks (GNNs) [8], [9] have made progress in modeling pedestrian movements and interactions. Other efforts focus on stochastic prediction architectures tailored to generate multimodal outputs, e.g., conditional variational auto-encoders (CVAEs) [10] or generative adversarial networks (GANs) [11]. Nevertheless, these studies mainly support an "one-size-fits-all" scheme that attempts to train a uniform pedestrian prediction model for various scenarios, ignoring the differences between trajectory domains. Once tested on new scenes, these methods tend to be less effective due to the over-specialization on training scenes.

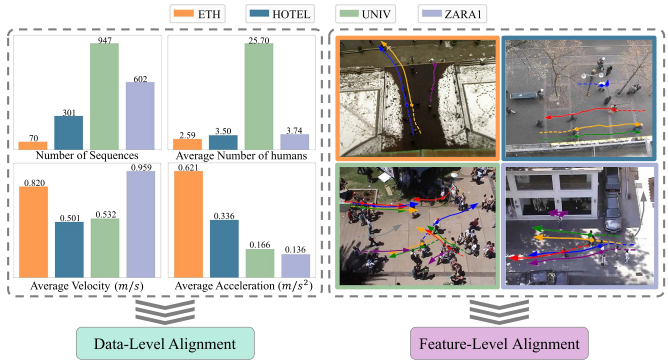


Fig. 1. Trajectory domain differences of four scenes: ETH, HOTEL, UNIV, and ZARA1. The left half shows macro differences, which are alleviated by our data-level alignment module. The right half shows micro differences, which are alleviated by our feature-level alignment module.

We argue that trajectory domain differences can be analyzed at macro and micro scales. As shown in Fig. 1, the macroscopic domain differences include statistical attributes such as average pedestrian number, velocity, and acceleration (e.g., the average velocity of ZARA1 is almost twice that of UNIV). The microscopic domain differences include obstacle distributions and individual walking habits (e.g., people prefer walking straight in parallel on leisure streets in ZARA1 but turning to avoid collision frequently in UNIV). We detail these macro and micro differences in Section III-B1 and III-C1.

Recent works attempt to introduce domain adaptation (DA) to solve this cross-domain prediction problem. T-GNN [12] employs an intermediate module to learn domain-invariant trajectory features. AML [13] adopts a post-calibration strategy which takes effect only at the inference phase. And BMP [14] assigns a physical motion prior to enhance the domain generalization

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Code is available at *https://github.com/jackyzengl/DADA*.

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ability. However, these existing methods neglect the macro and micro scales of cross-domain differences while only focus on single level for DA prediction.

To overcome the aforementioned limitations, we seek to design a multi-level domain transfer structure that can fulfill alignment at data level and feature level. It can be divided into two parts: data-level alignment aims to initially alleviate macroscopic differences and generate coarse-grained trajectory data with target-domain style; feature-level alignment is used to furthermore explore microscopic domain-invariant knowledge to reduce the prediction error.

In this letter, we propose a two-stage transferable architecture for pedestrian trajectory prediction. Firstly, we propose a novel data-level alignment model—a GAN constructed by temporal convolutional networks (TCNs), whose generator includes a graph attention module for extracting crowd social interactions. And we introduce a cycle-consistency loss to improve its stability [15]. Secondly, we propose a new feature-level alignment model—an auxiliary adversarial network, which assists in enhancing the backbone network's cross-domain robustness, thereby alleviating the micro differences elegantly. The key contributions of our work can be summarized as follows:

- We present a novel dual-alignment adaptive framework that achieves effective cross-domain knowledge transfer at both data level and feature level, which can be applied to public Encoder-Decoder prediction models.
- We propose a specifically designed temporal convolutional generative network for data-level alignment and an auxiliary adversarial network for feature-level alignment to improve model's cross-domain prediction ability.
- The experiments confirm that existing trajectory prediction networks integrated with our method achieve noticeable performance improvements, one of which shows the state-of-the-art performance and outperforms previous domain adaptation models.

II. Related Works

A. Human Trajectory Forecasting

The purpose of pedestrian trajectory prediction is to anticipate the future location of people based on their past paths. Modeling pedestrian movements and human-human interactions is the key to predicting reasonable trajectories. Pioneering works attempt to simulate pedestrian behaviors through mathematical statistics [16], optimal estimations [17] or artificial rules [18], which are limited in adapting to complicated scenarios. Recently, data-driven approaches have been moving toward the mainstream.

Due to the inherent timing characteristics of movement, Social-LSTM [5] firstly treats trajectory prediction as a sequence-to-sequence generation task, and utilizes Recurrent Neural Networks (RNNs) to extract temporal features. Follow-up works [6], [7] optimize the architecture of RNNs to refine feature quality. As Transformer [19] has gained prominence in Natural Language Processing field, some studies [20], [21] employ attention mechanism to predict future locations.

Capturing social and spatial relationships among pedestrians is also crucial in trajectory prediction tasks. Early works expect to model social interactions through the social pooling mechanisms [5], [11]. Recently, some approach [22] construct the entire scenario as a graph. Some work [8] based on Graph Attention Networks (GATs) [23] assign higher weights to edges between pairs of nodes with higher social affinity.

To generate diverse plausible prediction trajectories, Generative Adversarial Networks (GANs) and Conditional Variational Auto-Encoders (CVAEs) are introduced in recent years. These methods typically derive a latent variable from the observed trajectory, and then generate multi-modal outputs through a generative framework [24] or a learnable random sampling process [22].

Current approaches tend to be diversified, i.e., motion equivariance [25], diffusion model [26], and memory network [27]. In summary, existing methods mainly assume that there is a uniform motion pattern to learn, but seldom pay attention to the scene differences between different datasets. Our work focuses on narrowing the gap between training and testing scenes, which ensures the reliability in real-world applications.

B. Generative Adversarial Networks

GAN-based methodology has demonstrated excellent performance in image inpainting [28] and image-to-image translation [15]. The basic idea is to adopt an adversarial loss that makes the generated fake samples indistinguishable from the real ones. Some trajectory prediction methods [11], [24] also introduce GAN, regarding the predicted paths as fake samples and the ground-truth future paths as real samples on training, which all originate from the same datasets. Different from these works that simply perform trajectory generation in a domain, we employ a new Temporal Convolutional Networks (TCNs) based GAN for learning transferable knowledge between the source and target domain, which improves cross-domain performance.

C. Domain Adaptation

Domain adaptation (DA) [29] aims to diminish the discrepancy from source to target domains. In general, strategies of DA can fall into two categories: feature-level alignment model and generative model. The former reduces inter-domain divergences via minimizing the predefined statistical distance of feature distribution [30], whereas the latter takes the generated target domain samples as a kind of data augmentation, carries out domain alignment at pixel and feature levels [31].

Since trajectory datasets are unlabelled, T-GNN [12] treat trajectory DA as an unsupervised task and chooses 2-norm distance to align the domains. AML [13] tackle this problem in a test-time training way. BMP [14] use handcrafted motion priors to solve it. We adopt the generative model strategy for data-level alignment and utilize the adversarial training strategy for feature-level alignment to improve the expressiveness of the source-domain trained model on the target domain.

III. Proposed Method

The overall framework of our approach is illustrated in Fig. 2, which falls into two main phases: 1) a GAN based on TCNs to preliminarily align the information in the source and target domains at data level; 2) an adversarial network to promote the

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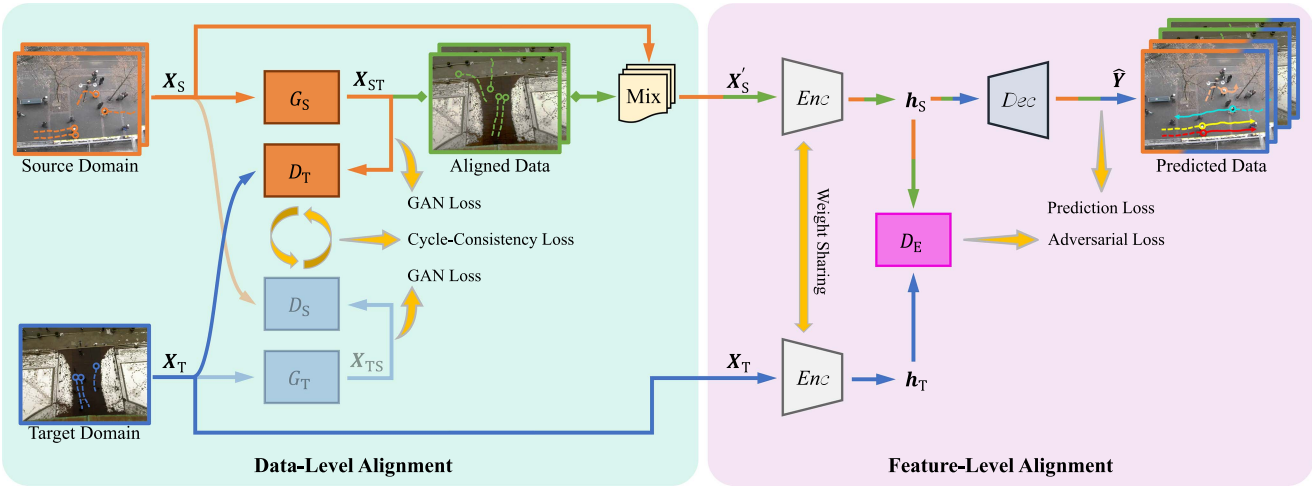


Fig. 2. The framework of our method during the **training phase**. Given the source and target domain trajectory inputs X_{S} and X_{T}, we construct two generators G_{S}, G_{T}, and two discriminators D_{S}, D_{T} based on TCNs to accomplish data-level alignment. This part is co-trained by GAN loss and cycle-consistency loss. Then, the X_{ST} generated by G_{S} is mixed with X_{S} to obtain the augmented data X'_{S}. Afterward, X'_{S} and X_{T} are passed through the encoder Enc to obtain the features h_{S} and h_{T} respectively, which are further fed to a discriminator D_{E} to complete the feature-level alignment. At last, the future prediction trajectory $\hat{Y}$ is derived from h_{S} via decoder Dec. This part is trained jointly by prediction loss and adversarial loss. During the **inference phase**, the test data of target domain is directly passed through well-trained Enc and Dec to output the prediction trajectory $\hat{Y}$, which could be used for downstream tasks in applications.

backbone encoder to further learn domain transferable features and improve the generalization performance. In Section III-B and III-C, we meticulously analyze the trajectory domain discrepancy problem and elaborate on our specific scheme.

A. Problem Definition

The pedestrian trajectory prediction task is formulated as one that infers the future trajectories of each pedestrian based on his/her historical observations. To be specific, we assume that there are N pedestrians in the scene and observe their positions in t_{obs} time-frames. The observed trajectory sequence is denoted as $X_i = \{X_i^t | t = 1, 2, \ldots, t_{\mathrm{obs}}\}$ for $\forall i \in \{1, 2, \ldots, N\}$, where $X_i^t = (x_i^t, y_i^t)$ is the coordinate of pedestrian i at t time-frame. Similarly, the predicted future trajectory is defined as $\hat{Y}_i = \{\hat{Y}_i^t | t = t_{\mathrm{obs}} + 1, t_{\mathrm{obs}} + 2, \ldots, t_{\mathrm{pred}}\}$ where $\hat{Y}_i^t = (\hat{x}_i^t, \hat{y}_i^t)$. Our goal is to build a prediction model with X as input, such that its output $\hat{Y}$ approximates the ground-truth Y as closely as possible.

For the cross-domain trajectory prediction task, the prediction model is firstly trained on the source domain data X_{S} and Y_{S}. Then, in evaluation phase, the model is deployed to a new target domain, where it generates the output $\hat{Y}_{\mathrm{T}}$ using X_{T} as input. A smaller discrepancy between $\hat{Y}_{\mathrm{T}}$ and Y_{T} indicates a better domain-adaption ability of the model.

B. Data-Level Alignment (DLA)

1) Macroscopic Domain Differences:: Objectively, there are massive macroscopic differences within diverse scenes. Busy university plazas and laid-back town streets differ in the flow and density of pedestrians, which leads to different kinds of social interactions in crowds and different collective motions in statistical properties, such as the average movement velocity and acceleration [12].

Above disparities appear at the data level, and it is difficult to reduce domain gaps using naive statistical methods. Intuitively, GAN is well suited for solving this problem as a transfer learning model widely used in computer vision [15]. We thus propose a GAN-structured network to accomplish the data-level alignment.

2) Attention-Based TCNs for GAN: For solving the domain transfer problem, we introduce a GAN with two parts: a generative model G that learns the distribution of target domain data, and a discriminative model D that distinguishes target domain data from G-generated data. Considering that trajectory data are essentially time series, there exists a high degree of temporal correlation. Compared with the frame-by-frame processing RNNs, the convolution operation is more capable of aggregating motion features [32] in adjacent periods, avoiding the errors accumulated over time [9]. Also inspired by the success of convolutional networks in previous works on style transfer [33], we employ temporal convolutional networks (TCNs) to form G and D to obtain the temporal information in the original data.

Generator G is in charge of transforming the data of source domain into the data of target domain. Given input trajectory sequence $X \in \mathbb{R}^{N \times 2 \times t_{\mathrm{obs}}}$, we perform successive convolution operations in its time dimension and expect the output sequence with target domain characteristics. The n-th layer of TCNs can be formally expressed as:

$$c^{(n)} = TCN(c^{(n-1)}; \mathbf{W}_{\mathrm{g}}^{(n)}), \tag{1}$$

where $c^{(n)} \in \mathbb{R}^{N \times F^{(n)} \times t_{\mathrm{obs}}}$ represents the output of the n-th layer (F is the feature dimension), $c^{(0)} = X$, and $\mathbf{W}_{\mathrm{g}}$ represent the learnable parameters in this generator. We find that 4-layer TCNs work best, and the final output is written as $G(X) \in \mathbb{R}^{N \times 2 \times t_{\mathrm{obs}}}$.

Discriminator D is employed to judge the truth of $G(X)$ and derive the corresponding true/false classification scores. Similar

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to the generator, the discriminator also consists of four layers of TCNs, and its final output is denoted as:

$$D(G(X)) = TCNs(G(X); \mathbf{W}_{\mathrm{d}}), \tag{2}$$

where $D(G(X)) \in \mathbb{R}^{N \times t_{\mathrm{obs}}}$ represents the classification score of $G(X)$, and $\mathbf{W}_d$ represent the learnable parameters in discriminator.

Taking into account that there are more complex social interactions between pedestrians that affect the social acceptability of the generated trajectories [8], we include an attention module in the generator to extract the social and spatial relationships between people. Specifically, we inserted a GAT [23] layer between the second and third layer of TCNs, denoted as:

$$c_{\mathrm{attn}}^{(2)} = GAT(c^{(2)}; \mathbf{W}_{\mathrm{gat}}), \tag{3}$$

where $c_{\mathrm{attn}}^{(2)}$ replaces $c^{(2)}$ as the input to the third layer of TCNs, and $\mathbf{W}_{\mathrm{gat}}$ are the learnable parameters of the GAT layer. Excessive layers of graph attention can lead to an overpowered generator, which is not conducive to overall training. In our implementation, one layer of GAT is used for the best results.

3) Objective Functions: For a task that transfers data from a source domain S to a target domain T, the generator is G_{S} and the discriminator is D_{T}. The training process is a min-max two-player game between these two, which eventually converges into a Nash equilibrium. We denote its loss function as:

$$\begin{aligned}\mathcal{L}_{\mathrm{GAN}}(G_{\mathrm{S}}, D_{\mathrm{T}}) = \ & \mathbb{E}_{X_{\mathrm{T}} \sim p_{data}(X_{\mathrm{T}})}[\log D_{\mathrm{T}}(X_{\mathrm{T}})] \\ & + \mathbb{E}_{X_{\mathrm{S}} \sim p_{data}(X_{\mathrm{S}})}[\log (1 - D_{\mathrm{T}}(G_{\mathrm{S}}(X_{\mathrm{S}})))],\end{aligned} \tag{4}$$

where p_{data} represents the data distribution, and $X_{\mathrm{S}}, X_{\mathrm{T}} \in \mathbb{R}^{N \times 2 \times t_{\mathrm{obs}}}$ are the samples from the source and target domains.

Considering that there are other alignment operations to follow, we must ensure that the data level alignment is stable. In other words, we aim to make the aligned data have general properties of target domain while retaining some specific knowledge of source domain, to diversify the model output rather than inferring similar distributions for conservative reasons during training. Inspired by previous work [15], we adopt a cycle-consistency loss to address this issue.

Subsequently, we add a generator G_{T} for restoring the transformed data to source domain, and a discriminator D_{S} for measuring the amount of source-domain knowledge retained therein. Similar to (4), G_{T} and D_{S} need to game each other to get reinforcement together, thus there is also an optimization goal: $\mathcal{L}_{\mathrm{GAN}}(G_{\mathrm{T}}, D_{\mathrm{S}})$.

In view of the fact that cycle-consistency needs to be satisfied in both the forward and backward directions, we denote cycle-consistency loss as:

$$\begin{aligned}\mathcal{L}_{\mathrm{cyc}}(G_{\mathrm{S}}, G_{\mathrm{T}}) = \ & \mathbb{E}_{X_{\mathrm{S}} \sim p_{data}(X_{\mathrm{S}})}[diff(X_{\mathrm{S}}, G_{\mathrm{T}}(G_{\mathrm{S}}(X_{\mathrm{S}})))] \\ & + \mathbb{E}_{X_{\mathrm{T}} \sim p_{data}(X_{\mathrm{T}})}[diff(X_{\mathrm{T}}, G_{\mathrm{S}}(G_{\mathrm{T}}(X_{\mathrm{T}})))],\end{aligned} \tag{5}$$

where $diff(\cdot)$ represents a function that calculates the difference between two inputs $X_{\mathrm{A}}, X_{\mathrm{B}} \in \mathbb{R}^{N \times F \times L}$ (L is the temporal dimension). In practice, Mean Squared Error (MSE) is used:

$$diff(X_{\mathrm{A}}, X_{\mathrm{B}}) = \frac{1}{NL} \sum_{i=1}^{N} \| {X_{\mathrm{A}}}_i^{1:L} - {X_{\mathrm{B}}}_i^{1:L} \|_2^{\ 2}. \tag{6}$$

In summary, overall objective for data-level alignment is:

$$\begin{aligned}\mathcal{L}_{\mathrm{data}} = \ & \mathcal{L}_{\mathrm{GAN}}(G_{\mathrm{S}}, D_{\mathrm{T}}) + \mathcal{L}_{\mathrm{GAN}}(G_{\mathrm{T}}, D_{\mathrm{S}}) \\ & + \ \alpha \, \mathcal{L}_{\mathrm{cyc}}(G_{\mathrm{S}}, G_{\mathrm{T}}),\end{aligned} \tag{7}$$

where α is the weight factor of the cycle-consistency loss. Our aim is to obtain the optimal generative model G_{S}^* by jointly training the four networks:

$$G_{\mathrm{S}}^* = \arg \min_{G_{\mathrm{S}}, G_{\mathrm{T}}} \max_{D_{\mathrm{S}}, D_{\mathrm{T}}} \mathcal{L}_{\mathrm{data}}(G_{\mathrm{S}}, G_{\mathrm{T}}, D_{\mathrm{S}}, D_{\mathrm{T}}), \tag{8}$$

to eventually yield the aligned data $X_{\mathrm{ST}}^* = G_{\mathrm{S}}^*(X_{\mathrm{S}})$. To further ensure the stability of this process, generated data X_{ST}^* will be mixed with the original dataset of X_{S} to produce the augmented source-domain data X_{S}'.

C. Feature-Level Alignment (FLA)

1) Microscopic Domain Differences: Various scenes are also divergent at the micro level. For one scene, the walking habits of pedestrians of all ages naturally diversify, and the movement trajectories of pedestrians with disparate destinations (far or near) also stylistically differ. Meanwhile, obstacle distribution is an inner characteristic of a scene which may influence human's choices of future paths.

Since microscopic features are typically captured on a high-dimensional feature space, mere alignment at the data level is not sufficient and requires further processing at the feature level. We consider the source-to-target microscopic disparities as broad-sense "adversarial noise", and the macro-aligned source samples (X_{S}') as target-domain "adversarial examples" mixed with this kind of noise. Inspired by adversarial training [34], we introduce a discriminator for adversarial attacking training to assist the prediction model in accommodating this "adversarial noise", thereby accomplishing the feature-level alignment.

2) Embedded Discriminator: We observe that most recent trajectory prediction models are in encoder-decoder structure, and therefore embed a discriminator between the encoder and decoder of the backbone for domain alignment at the intermediate feature level. Specifically, we take the augmented source-domain data X_{S}' and target-domain data X_{T} as inputs, and their respective encoder outputs as the features used for alignment:

$$\begin{cases} h_{\mathrm{S}} = Enc(X_{\mathrm{S}}'; \mathbf{W}_{\mathrm{enc}}), \\ h_{\mathrm{T}} = Enc(X_{\mathrm{T}}; \mathbf{W}_{\mathrm{enc}}), \end{cases} \tag{9}$$

where $h \in \mathbb{R}^{N \times F}$ presents the features to be aligned, Enc denotes the encoder of backbone, and $\mathbf{W}_{\mathrm{enc}}$ are its learnable parameters.

Then, an embedded discriminator D_{E} further distinguishes whether the features are from the target domain or not, and infers the relevant confidence scores. The discriminator consists of several MLP layers attached ReLU activation layers.

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3) Objective Functions: Once the feature h_{S} is obtained, the final prediction output can be derived via the decoder: $\hat{Y} = Dec(h_{\mathrm{S}})$. Different backbones have diverse objectives to optimize the prediction process. For convenience, we denote the prediction loss uniformly as:

$$\mathcal{L}_{\mathrm{pred}}(Enc, Dec) = loss(\hat{Y}, Y), \tag{10}$$

where $\hat{Y} \in \mathbb{R}^{N\times 2\times t_{\mathrm{pred}}}$ represents the predicted trajectories, Y represents the ground-truth, and $loss(\cdot)$ is the prediction loss function the backbone adopts.

Besides, we employ an adversarial loss to jointly train the encoder Enc and D_{E} to accomplish feature-level domain alignment as follows:

$$\mathcal{L}_{\mathrm{adv}}(Enc, D_{\mathrm{E}}) = \mathbb{E}_{\substack{h_{\mathrm{S}}\sim p_{data}(h_{\mathrm{S}}) \\ h_{\mathrm{T}}\sim p_{data}(h_{\mathrm{T}})}}[diff(D_{\mathrm{E}}(h_{\mathrm{S}}), D_{\mathrm{E}}(h_{\mathrm{T}}))]. \tag{11}$$

Therefore, the total loss of feature-level alignment is:

$$\begin{aligned}\mathcal{L}_{\mathrm{feat}}(Enc, Dec, D_{\mathrm{E}}) = \ & \mathcal{L}_{\mathrm{pred}}(Enc, Dec) \\ & + \beta\, \mathcal{L}_{\mathrm{adv}}(Enc, D_{\mathrm{E}}),\end{aligned} \tag{12}$$

where β is a hyperparameter to balancing the scale.

We attempt to optimize the encoder to capture domain-invariant knowledge at feature level via following procedure:

$$Enc^{*} = \arg \min_{Enc, Dec} \max_{D_{\mathrm{E}}} \mathcal{L}_{\mathrm{feat}}(Enc, Dec, D_{\mathrm{E}}). \tag{13}$$

IV. Experiments

In this section, we conduct comprehensive experiments on public benchmark datasets following the evaluation protocol of T-GNN [12], to verify how our domain adaptation approach contributes to pedestrian trajectory prediction. We first state our experimental setup (Section IV-A), then integrate our DLA and FLA modules into conventional baselines to demonstrate the effectiveness of each component of our approach (Section IV-B). And we evaluate our method in comparison with existing state-of-the-art domain adaptation methods (Section IV-C). In addition, we conduct ablation studies of the GAT module and two hyperparameters in our method (Section IV-D). Finally we provide qualitative results to illustrate the benefits of our approach (Section IV-E).

A. Experimental Setup

1) Datasets: We evaluate the performance of our method on two widely used public datasets over previous methods: ETH [35] and UCY [36]. The ETH and UCY datasets consist of real-world human trajectories with complex social interactions, divided into scenes—ETH and HOTEL, and UNIV, ZARA1 and ZARA2, respectively. These five scenes differ in geographic area as well as in scenario layout.

2) Task Setting: Each scene from the ETH and UCY datasets is treated as a separate trajectory domain. The models are trained on a source domain S and evaluated on another target domain T. We term this task as "S2T". We denote ETH, HOTEL, UNIV, ZARA1, and ZARA2 scenes by A, B, C, D, and E, respectively. Accordingly, there are total $P(5, 2) = 20$ tasks, represented as A2B, B2C, C2D, and so on.

3) Baselines: We compare the proposed approach with state-of-the-art models, which come in two categories: 1) conventional baselines without any adaptation measures: **S-GAN** [11], **STGAT** [8], **S-STGCNN** [9], **PECNet** [10], **Trajectron++** [23], **TUTR** [21]; 2) DA models, i.e., recently proposed trajectory prediction methods with domain adaptation modules: **T-GNN** [12], **AML** [13], **BMP** [14].

4) Evaluation Protocols: To achieve comparative fairness, we apply different evaluation protocols for different categories of models, as in recent work [12]. Take task S2T as an example, for the conventional baselines, the source dataset's training set $\mathbb{D}_{\mathrm{S}}^{\mathrm{train}}$ and the target dataset's validation set $\mathbb{D}_{\mathrm{T}}^{\mathrm{val}}$ are used for training, while the target dataset's test set $\mathbb{D}_{\mathrm{T}}^{\mathrm{test}}$ is used for evaluation. As for the DA models, we take $\mathbb{D}_{\mathrm{S}}^{\mathrm{train}}$ as the source domain and $\mathbb{D}_{\mathrm{T}}^{\mathrm{val}}$ as the target domain for adaptive training, and $\mathbb{D}_{\mathrm{T}}^{\mathrm{test}}$ is only available for evaluation. There is no overlap between $\mathbb{D}_{\mathrm{T}}^{\mathrm{val}}$ and $\mathbb{D}_{\mathrm{T}}^{\mathrm{test}}$. Our approach follows the evaluation protocol for DA models. We observe trajectories for 3.2 seconds (8 frames), then predict paths for the next 4.8 seconds (12 frames).

5) Metrics: We evaluate the performance of each trajectory prediction model using two metrics. *Average Displacement Error* (ADE) represents the mean square error between the predicted trajectory and the ground truth trajectory of all pedestrians at all frames. *Final Displacement Error* (FDE) refers to the average distance between the predicted final destination and the true final destination of all pedestrians at the last frame. The lower these metrics are reported, the better the prediction performance of methods.

6) Experimental Platform: We adopt 1 NVIDIA 24G 4090 GPU for training. As for DLA generative model, we use Adam optimizer with learning rate, batch size and epoch set to 0.0005, 32 and 250. As for FLA module, the learning rate and grad clipping norm are set to 0.001 and 1.5. As for prediction model, all training settings follow the official baseline.

B. Performance Improvement on Baselines

In this experiment, our proposed DLA serves as a callable module at the input of the models during the training phase, providing aligned source domain data. Our FLA serves as an embedded module to align the intermediate features, enabling the encoder of models to learn cross-scene knowledge.

We integrate the DLA, FLA and DADA (including DLA and FLA) modules individually into six conventional baseline models, and report the FDE results on 20 tasks in Table I. With the application of DLA and FLA module, all six baselines yield great accuracy gains on most tasks, which is also reflected in the consistent reduction of average FDE metric for all baselines (up to 53.5%). In addition, we find that our approach has limited improvement on some C2X (X represents a certain target scene) tasks. It may be because that UNIV dataset has the greatest number of samples, which cover multiple pedestrian movement patterns. Similarly, for tasks D2E and E2D, since D and E have similar scenario layouts, the domain adaptation effect is theoretically less prominent.

Finally, to explore how much remaining room is for future improvement, these conventional models are trained with target

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TABLE I
FDE METRIC COMPARISON RESULTS OF BL (BASELINE), DLA, FLA, DADA (DLA + FLA) AND IDEAL FOR SIX CONVENTIONAL MODELS ON 20 TASKS. A, B, C, D, AND E DENOTE ETH, HOTEL, UNIV, ZARA1, AND ZARA2. "2": FROM SOURCE DOMAIN TO TARGET DOMAIN. GAIN↑: AVERAGE PERFORMANCE IMPROVEMENT W.R.T. BASELINE. UNIT FOR FDE: METER. **BOLD**: BEST AMONG BL, DLA, FLA, AND DADA

	S-GAN [11]					STGAT [8]					S-STGCNN [9]					PECNet [10]					Trajectron++ [22]					TUTR [21]				
	BL	DLA	FLA	DADA	IDEAL	BL	DLA	FLA	DADA	IDEAL	BL	DLA	FLA	DADA	IDEAL	BL	DLA	FLA	DADA	IDEAL	BL	DLA	FLA	DADA	IDEAL	BL	DLA	FLA	DADA	IDEAL
A2B	2.89	2.51	2.32	**1.03**	0.28	2.08	0.61	2.11	**0.59**	0.23	3.24	1.42	1.54	**1.01**	0.60	3.33	**0.32**	3.57	**0.32**	0.25	0.63	0.47	1.49	**0.35**	0.15	0.69	**0.23**	0.34	**0.23**	0.36
A2C	2.48	4.28	3.01	**2.38**	0.63	1.54	1.54	1.61	**1.49**	1.01	2.86	1.84	1.22	**0.98**	0.67	2.83	**1.43**	1.92	1.44	0.89	0.89	0.99	**0.88**	0.96	0.41	0.62	**0.48**	0.61	0.51	0.30
A2D	3.37	3.65	3.10	**1.85**	0.30	1.26	1.08	1.02	**1.01**	0.59	2.53	1.28	1.99	**1.10**	0.45	2.53	1.20	1.37	**1.08**	0.96	0.74	0.55	0.65	**0.54**	0.30	0.65	0.57	**0.55**	**0.55**	0.21
A2E	2.85	3.00	2.29	**1.36**	0.35	1.04	**0.88**	0.94	0.89	0.58	2.43	1.77	**1.52**	1.62	1.28	2.45	1.17	1.04	**1.03**	1.36	0.65	**0.46**	0.51	0.47	0.27	0.43	**0.37**	0.44	**0.37**	0.78
B2A	5.37	**2.18**	4.84	2.49	1.75	3.81	**1.11**	3.82	1.12	1.28	5.16	1.46	2.83	**1.12**	1.75	5.23	1.71	2.71	**1.69**	1.80	3.30	1.12	3.33	**0.99**	0.60	0.95	0.79	0.84	**0.73**	0.19
B2C	2.74	2.84	2.19	**2.08**	0.63	1.59	1.35	1.63	**1.33**	1.01	2.51	1.39	2.17	**1.23**	0.67	2.48	**1.77**	1.83	1.80	0.89	1.49	0.81	1.51	**0.65**	0.41	0.59	**0.47**	0.62	0.48	0.30
B2D	4.26	2.22	4.23	**1.45**	0.30	2.58	**1.00**	2.46	1.24	0.59	4.86	**1.10**	2.49	1.20	0.45	4.90	1.02	3.12	**0.98**	0.96	3.55	**0.67**	3.67	0.70	0.30	0.49	**0.47**	0.49	0.48	0.21
B2E	2.78	**1.79**	2.81	2.05	0.35	1.61	**0.85**	1.67	0.99	0.58	2.88	1.53	1.93	**1.24**	1.28	2.86	**0.90**	1.97	1.00	1.36	1.98	0.46	2.08	**0.45**	0.27	0.36	**0.34**	0.37	0.36	0.78
C2A	3.25	2.69	**2.02**	2.38	1.75	1.44	1.77	1.28	**1.07**	1.28	2.30	**0.99**	2.60	1.13	1.75	2.22	1.95	**1.71**	2.11	1.80	1.12	1.18	**0.87**	1.13	0.60	0.64	0.64	**0.63**	0.64	0.19
C2B	2.56	1.42	1.17	**1.13**	0.28	0.73	0.70	**0.67**	0.69	0.23	1.34	0.98	1.42	**0.60**	0.60	1.32	**0.56**	0.65	**0.56**	0.25	0.37	0.29	0.32	**0.27**	0.15	0.23	**0.21**	0.24	0.23	0.36
C2D	2.61	1.83	**1.13**	1.34	0.30	**0.70**	0.72	0.73	0.71	0.59	1.74	**0.75**	1.42	0.81	0.45	1.68	0.93	**0.73**	0.96	0.96	0.41	0.63	**0.40**	0.59	0.30	**0.37**	0.38	0.38	0.38	0.21
C2E	1.70	1.57	**0.88**	0.92	0.35	**0.60**	0.65	0.65	0.68	0.58	1.10	**0.59**	0.69	0.84	1.28	1.12	1.05	**0.65**	1.03	1.36	0.31	0.32	**0.30**	0.36	0.27	0.28	**0.27**	0.28	0.29	0.78
D2A	2.20	2.27	**2.16**	2.25	1.75	1.31	1.14	1.24	**1.02**	1.28	2.21	**1.19**	1.74	1.20	1.75	2.20	1.61	2.37	**1.52**	1.80	1.43	**0.89**	1.53	0.96	0.60	0.67	0.68	0.91	**0.61**	0.19
D2B	2.22	1.58	1.61	**1.02**	0.28	1.05	**0.48**	0.78	0.67	0.23	1.99	1.59	1.45	**1.56**	0.60	2.05	0.60	2.79	**0.48**	0.25	0.90	**0.31**	1.10	0.32	0.15	0.21	**0.18**	0.21	**0.18**	0.36
D2C	1.77	1.80	**1.41**	1.56	0.63	**1.16**	1.22	1.20	1.22	1.01	1.41	0.98	1.14	**0.95**	0.67	1.52	**1.43**	1.63	1.44	0.89	0.66	**0.65**	0.74	0.66	0.41	0.47	**0.44**	0.48	**0.44**	0.30
D2E	1.44	1.23	0.88	**0.70**	0.35	0.66	0.72	0.66	**0.63**	0.58	0.88	0.54	1.46	**0.53**	1.28	0.88	**0.83**	2.60	0.85	1.36	0.38	0.40	**0.35**	0.39	0.27	0.30	**0.28**	0.32	**0.28**	0.78
E2A	2.49	2.19	**1.96**	2.04	1.75	1.19	1.22	1.06	**0.95**	1.28	2.10	1.03	1.89	**1.00**	1.75	2.10	1.86	2.41	**1.26**	1.80	1.29	1.16	1.43	**0.97**	0.60	0.76	0.70	0.72	**0.69**	0.19
E2B	2.32	2.03	1.89	**1.46**	0.28	0.94	**0.79**	0.89	0.94	0.23	2.05	1.67	1.64	**1.47**	0.60	1.84	1.15	1.54	**0.77**	0.25	0.81	0.36	1.10	**0.26**	0.15	0.19	0.19	0.19	0.19	0.36
E2C	2.10	1.67	**1.54**	1.59	0.63	**1.19**	1.33	1.17	1.27	1.01	1.47	**0.92**	1.11	1.01	0.67	1.45	1.50	**1.35**	1.56	0.89	0.75	0.66	0.80	**0.63**	0.41	0.45	**0.44**	0.46	0.45	0.30
E2D	1.25	1.35	1.03	**0.94**	0.30	0.77	**0.72**	0.75	0.79	0.59	1.01	**0.78**	1.32	**0.78**	0.45	0.98	1.17	**0.89**	1.07	0.96	**0.41**	0.52	0.50	0.52	0.30	**0.36**	0.37	0.37	0.38	0.21
AVG	2.63	2.21	2.12	**1.60**	0.66	1.36	0.99	1.31	**0.96**	0.67	2.30	1.19	1.68	**1.07**	0.73	2.29	1.21	1.84	**1.15**	1.05	1.10	0.65	0.99	**0.61**	0.37	0.48	0.43	0.46	**0.42**	0.37
Gain↑	-	16.0%	19.4%	39.2%	-	-	27.2%	3.2%	29.4%	-	-	48.3%	27.0%	53.5%	-	-	47.2%	19.7%	50.0%	-	-	40.9%	10.0%	44.5%	-	-	10.4%	4.2%	12.5%	-

TABLE II
FDE METRIC COMPARISON RESULTS BETWEEN DADA-MODELS AND OTHER DA MODELS. **BOLD**: BEST, UNDERLINE: SECOND BEST. "TRAJ" AND "TUTR" REPRESENT TRAJECTRON++ [22] AND TUTR [21] AS BACKBONE

Methods	Performance (FDE)↓ (Source2Target)																				AVG
	A2B	A2C	A2D	A2E	B2A	B2C	B2D	B2E	C2A	C2B	C2D	C2E	D2A	D2B	D2C	D2E	E2A	E2B	E2C	E2D	
T-GNN [12]	2.18	2.25	1.78	1.84	4.15	1.82	4.04	2.53	1.91	1.12	1.30	0.87	1.92	1.46	1.25	0.65	1.86	1.45	1.28	0.72	1.82
AML [13]	0.64	1.19	1.06	0.81	1.65	1.25	0.92	0.69	1.97	0.81	0.93	0.84	1.80	0.56	1.01	0.69	1.75	0.63	1.05	0.93	1.06
BMP [14]	0.41	0.52	0.56	0.50	1.71	0.60	0.53	0.46	1.48	0.32	0.53	0.44	1.65	0.34	0.66	0.47	1.66	0.32	0.67	0.53	0.72
DADA-Traj	0.35	0.96	**0.54**	0.47	0.99	0.65	0.70	0.45	1.13	0.27	0.59	0.36	0.96	0.32	0.66	0.39	0.97	0.26	0.63	0.52	0.61
DADA-TUTR	**0.23**	**0.51**	0.55	**0.37**	**0.73**	**0.48**	**0.48**	**0.36**	**0.64**	**0.23**	**0.38**	**0.29**	**0.61**	**0.18**	**0.44**	**0.28**	**0.69**	**0.19**	**0.45**	**0.38**	**0.42**

domain training set and then evaluated with target domain test set. We refer to the results as IDEAL in Table I, which indicates that our method narrowed the IDEAL gap by 68.9% on average compared to the baseline model without any domain adaptation operations. However, there is still large remaining room for improvement for six conventional models on most tasks.

C. Comparison With Other DA Methods

We report in Table II the FDE results of our proposed method compared with three state-of-the-art DA models. To ensure fairness in the comparison, we choose Trajectron++ and TUTR as backbone, which were proposed before and after the three DA models, respectively. In general, DADA-models outperform the three DA models on all tasks, with up to 41.7% improvement on average FDE. In contrast to T-GNN, we employ coarse-grained domain alignment at the data level. Then, rather than using a single inter-domain distance loss, an adversarial loss is used to dynamically align feature-level information to further correct the shifts in the high-dimensional feature space. As compared to AML, we first choose the input side where spatio-temporal features of trajectories are richer, to implement partial scene information transfer. Then, tuning of the backbone network is completed by an intermediate-level auxiliary adaptation module, instead of additionally introducing a post-processing module at the output side. Differing from BMP, we focus on more holistic data-level scene differences and more latent feature-level knowledge on target domain, rather than individual-level motion priors.

TABLE III
ABLATION STUDY OF THE PERFORMANCE OF GRAPH ATTENTION MODULE (GAT) ON SOME TASKS. DLA*: DATA-LEVEL ALIGNMENT WITHOUT GAT MODULE. **BOLD**: BEST. BACKBONE: S-GAN [11]

Variants	DLA*	FLA	GAT	Performance (ADE / FDE)↓				
				A2B	B2C	C2D	D2E	E2A
DLA w/o GAT	✓			1.62 / 2.72	1.56 / **2.81**	0.88 / 1.92	0.64 / 1.46	1.24 / 2.60
DLA w/ GAT	✓		✓	**1.51 / 2.51**	**1.49** / 2.84	**0.80 / 1.83**	**0.55 / 1.23**	**0.97 / 2.19**
DADA w/o GAT	✓	✓		0.90 / 1.67	1.20 / 2.21	0.97 / 2.12	0.52 / 1.39	1.13 / 2.54
DADA w/ GAT	✓	✓	✓	**0.50 / 1.03**	**0.93 / 2.08**	**0.62 / 1.34**	**0.32 / 0.70**	**0.95 / 2.04**

D. Ablation Study

1) Effectiveness of Graph Attention Module: As mentioned in (3), we insert a graph attention module (GAT) between the second and third layer of TCNs in DLA. To analyze the contribution of GAT, we design an ablation study, as shown in Table III. It is performed on five tasks that consist of cross-scenes with large differences, which could be well used to explore the effectiveness of the individual components.

Four variants are generated depending on the adoption of components: 1) "DLA w/o GAT": a variant of DLA, where the graph attention module was removed, is applied on backbone; 2) "DLA w/ GA": a GAT module is inserted into the DLA of "DLA w/o GAT"; 3) "DADA w/o GAT": a FLA module is added based on "DLA w/o GAT"; 4) "DADA w/ GAT": an approach using all components. It indicates using GAT module results in a huge performance improvement on most tasks.

2) Performance Study of α in DLA: The hyperparameter α is used to control the relative importance of the cycle-consistency

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TABLE IV
ABLATION STUDY OF α WITH FIVE DIFFERENT VALUES. S-GAN [11] IS CHOSEN AS BACKBONE. **BOLD**: BEST. UNDERLINE: SECOND BEST

α	Performance (ADE / FDE)↓					AVG
	A2B	B2C	C2D	D2E	E2A	
0.01	1.41 / 2.40	1.19 / **2.24**	1.16 / 2.76	0.61 / 1.33	1.96 / 3.88	1.15 / 2.31
0.1	**1.17** / **2.11**	1.31 / 2.48	0.86 / 1.94	0.61 / 1.39	1.12 / **2.15**	1.16 / 2.30
1	1.51 / 2.51	1.49 / 2.84	**0.80** / **1.83**	0.55 / 1.23	**0.97** / 2.19	**1.11** / **2.21**
10	2.02 / 3.32	**1.15** / 2.28	0.98 / 2.25	**0.53** / **1.14**	1.25 / 2.45	1.14 / 2.23
100	1.55 / 2.63	1.31 / 2.43	0.87 / 1.98	0.73 / 1.62	1.07 / 2.39	1.14 / 2.26

TABLE V
ABLATION STUDY OF β WITH FIVE DIFFERENT VALUES. S-GAN [11] IS CHOSEN AS BACKBONE. **BOLD**: BEST. UNDERLINE: SECOND BEST

β	Performance (ADE / FDE)↓					AVG
	A2B	B2C	C2D	D2E	E2A	
0.01	1.13 / 1.98	1.33 / 2.44	0.73 / 1.57	0.40 / 0.84	**0.87** / 1.84	1.03 / 2.04
0.1	0.77 / 1.38	1.32 / 2.45	0.70 / 1.61	0.49 / 0.87	0.88 / **1.73**	0.99 / 1.94
1	1.88 / 3.48	**0.83** / **1.73**	0.68 / 1.45	1.40 / 2.57	0.98 / 2.04	0.96 / 1.90
10	**0.50** / **1.03**	1.11 / 2.08	**0.62** / **1.34**	**0.32** / **0.70**	0.95 / 2.04	**0.80** / **1.60**
100	0.75 / 1.33	0.84 / 1.75	2.50 / 4.59	0.33 / 0.71	0.97 / 2.07	0.84 / 1.68

loss in (7). We set different values $\alpha = \{0.01, 0.1, 1, 10, 100\}$ to provide the augmented source domain data with different alignment levels, which is further fed to S-GAN [11] as backbone to see how α affects the experimental results, as shown in Table IV. We adopt $\alpha = 1$ in our implementation, which performs best in average.

3) Performance Study of β in FLA: The hyperparameter β is used to balance the two terms in (12). Setting β too small leads to the lack of sufficient alignment, while setting β too large leads to the neglect of prediction loss. We set different values $\beta = \{0.01, 0.1, 1, 10, 100\}$ to find the most suitable value, as shown in Table V. In our implementation, $\beta = 10$ is adopted for the best average results.

E. Qualitative Analysis

1) Analysis of Predicted Trajectory Distribution: We select multiple scenarios and sample the predicted trajectory distributions for each pedestrian, as shown in Fig. 3. In scenario 1 and 2, following behavior occurs with parallel walking, which disturbs the baseline model from identifying forthcoming walking directions. However, DLA module helps indicate accurate directions, while DADA refines the trajectories to converge to the ground truth. Scenarios 3 and 4 show cases that pedestrians are going to collide while moving. In DLA of Scenario 3, the green individual is predicted to avoid collision with the red one by crossing paths, whereas the ground truth shows them moving apart. Similarly, in DLA of Scenario 4, the green individual is predicted to avoid the blue one by heading in reverse direction, which is contrary to the slightly directional adjustment in real. By introducing FLA, the DADA model predicts the correct trajectories with proper collision avoidance strategies. Scenario 5 represents a more complex example involving parallel walking, turning and collision avoidance. Even in case that the baseline fails, our DADA method generates socially plausible trajectories.

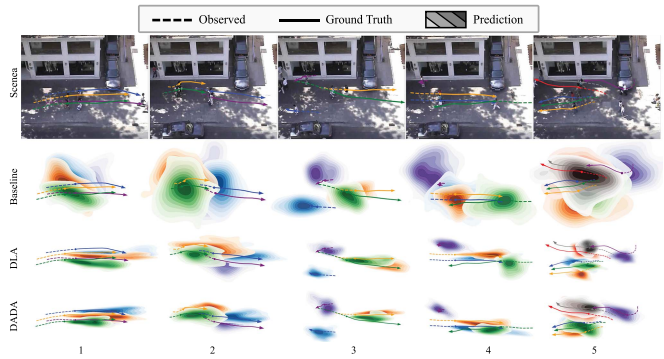


Fig. 3. Qualitative comparison of Baseline, DLA, and DADA (DLA + FLA) in five complex scenarios: (1) four individuals walking in parallel, (2) two people following the other two, (3) two people moving apart and the other two walking independently, (4) three people meeting from different directions and the other one standing still, (5) group walking and turning. Each color denotes a pedestrian, and the colored shading represents the multi-modal predicted trajectory distribution. The closer this distribution approximates GT (solid line), the higher the prediction accuracy. C2D is taken as the example task and S-GAN [11] is taken as the baseline.

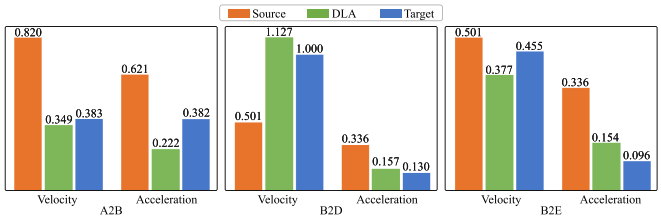


Fig. 4. Average velocity and acceleration statistics on three tasks. For each task, source and target data are obtained from the corresponding training set $\mathbb{D}_{\mathrm{S}}^{\mathrm{train}}$ and validation set $\mathbb{D}_{\mathrm{T}}^{\mathrm{val}}$, respectively. And DLA data is generated by our DLA module. Unit: m/s for velocity and m/s^2 for acceleration.

2) DLA Effectiveness Visualization: To objectively evaluate the macro domain gaps and the effectiveness of our DLA module, we give the statistical average velocity and acceleration in Fig. 4. It can be observed that the source data has huge motion disparities with the target, while our DLA data significantly aligns to the moving pattern of target domain and has more similar velocity and acceleration.

3) FLA Effectiveness Visualization: Micro domain disparities can also be analyzed by comparing the differences of intermediate feature representations—we provide visualizations of source domain features (h_{S}) and target domain features (h_{T}) with t-SNE approach. As shown in Fig. 5, by comparing the lower row with the upper row, we can conclude that FLA module alleviates scene disparities in two ways: for task A2B and B2D, there is an increased overlap between source and target distributions after applying FLA; for task B2E, FLA imposes a similar pattern constraint for both domain features.

V. CONCLUSION

In this letter, we work on solving the domain misalignment problem of pedestrian trajectory prediction, when the dataset for training and the dataset for real applications are from different scenarios. We propose a new two-stage domain alignment

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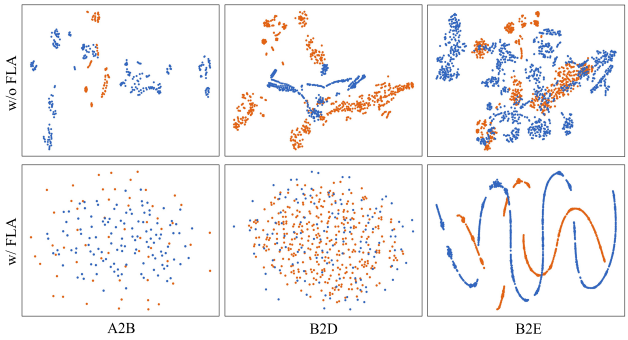


Fig. 5. The t-SNE visualization results of intermediate feature representations h_{S} and h_{T} on three tasks. Each dot represents the feature of a pedestrian. The red and blue dots denote the source and target features, respectively. "w/o FLA" denotes that all feature dots are extracted by the model without FLA. And "w/ FLA" denotes that feature dots are extracted by the model which is integrated with FLA during the training phase.

scheme to mitigate the differences between different domains, including DLA and FLA. The experiments verify the effectiveness of proposed method and its superiority over the previous conventional baselines and the recent DA methods.

Although our approach achieves surpasses performance over previous works, there is some room for improvement: 1) We consider it effective to automatically generate the α and β adaptive factors in Tables IV and V according to the cross-domain disparities; 2) It should be explored how to quantify various hierarchical features in the backbone network in order to align the layer with the greatest domain difference.

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